Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

DEPARTMENT OF MECHATRONICS

V SEMESTER B.TECH. (MECHATRONICS)

END SEMESTER EXAMINATIONS, [Dec] [2021]

SUBJECT: MACHINE VISION AND IMAGE PROCESSING

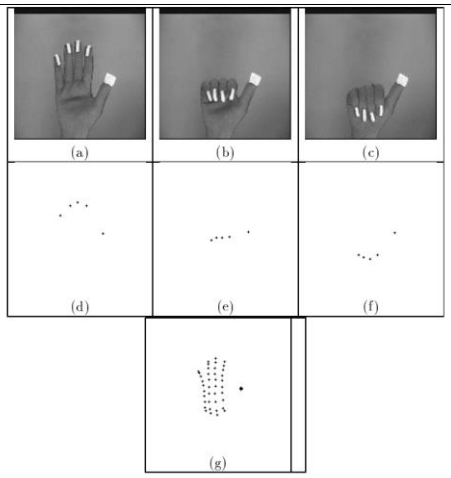
SUBJECT CODE: MTE 4075 **DATE:17/12/2021**

Time: 75+10 MINUTES **MAX. MARKS: 20**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data if any can be suitably assumed.

PART B

Q. No	Question	M	CO/ CLO	PO	LO	BL
1A.	 **Fig Q1A**. (a, b, c): Hand images, (d, e, f): End points of fingers, (g) Trajectory generation Implement the algorithm to select the end points of the fingers for the images shown in the **Fig Q1A (a, b, c)**. Propose the technique with detailed steps to generate the trajectory as shown in the **Fig Q1A. (g)**	**5**	**3,4**	**4**	**3,4**	**6**

1B.	Consider the image f as shown in the Fig Q1B. Assume that it is to be convolved with the mask m, to produce the output image g. Compute g9 and g41. f1 f2 f3 f4 f5 f6 f7 f8 f9 f10 f11 f12 f13 f14 f15 f16 f17 f18 f19 f20 f21 f22 f23 f24 f25 f26 f27 f28 f29 f30 f31 f32 f33 f34 f35 f36 f37 f38 f39 f40 f41 f42 f43 f44 f45 f46 f47 f48 f49 m1 m2 m3 m4 m5 m6 m7 m8 m9 g1 g2 g3 g4 g5 g6 g7 g8 g9 g10 g11 g12 g13 g14 g15 g16 g17 g18 g19 g20 g21 g22 g23 g24 g25 g26 g27 g28 g29 g30 g31 g32 g33 g34 g35 g36 g37 g38 g39 g40 g41 g42 g43 g44 g45 g46 g47 g48 g49 . Fig Q1B. f represents the input image, m represents the mask, g represents the output image.	**3**	**1**	**1**	**1,2**	**3**
1C.	Implement the algorithm to make a black and white Rubik's cube as a 3D joystick/mouse.	**2**	**1**	**4**	**1,2**	**5**
2A.	Elaborate the face recognition method using PCA with mathematical steps.	**5**	**4**	**4,5**	**6,12**	**5**
2B.	Identify two image denoising techniques based on the noise content. Does the performance of the algorithm depend on the correct choice of noise level estimate? Draw conclusions on the techniques which would work better for a particular type of noise?	**3**	**1**	**1**	**1**	**4, 5**
2C.	Implement the algorithm to replace a picture in a magazine or a book with a different image or video.	**2**	**1**	**4,5**	**2**	**6**